

Effort and Cost Estimation Techniques in Software Project Management

1. Introduction

In **Software Project Management (SPM)**, one of the most important tasks of a project manager is to estimate:

- **How much work is required?** → Effort estimation
- **How much time is required?** → Schedule estimation
- **How much money is required?** → Cost estimation

The basic relationship can be remembered as:

Software Size → Effort → Schedule → Cost

2. What is Effort Estimation?

Effort estimation is the process of predicting the amount of human work required to complete a software project.

Effort is generally measured in:

- Person-hours
- Person-days
- Person-months

Example

Suppose a project requires:

- 4 developers
- 5 months

Then approximate effort is:

$4 \times 5 = 20$ person-months

So, the estimated effort is **20 person-months**.

Important: Person-months measure effort, not necessarily calendar time.

3. What is Cost Estimation?

Cost estimation is the process of predicting the total financial resources required to complete a software project.

It may include:

- Employee salaries
- Hardware
- Software licenses
- Cloud infrastructure
- Testing
- Training
- Travel
- Communication
- Maintenance
- Contingency/reserve

Basic Formula

Total Cost=Labour Cost+Other Project Costs

4. Relationship Between Effort and Cost

Effort is one of the major inputs for calculating cost.

For example:

Estimated effort:

20 person-months

Average labour cost:

₹80,000 per person-month

Then:

$$20 \times ₹80,000 = ₹16,00,000$$

So labour cost = **₹16 lakh.**

If other costs are ₹4 lakh:

$$₹16L + ₹4L = ₹20L$$

Therefore:

Total estimated project cost = ₹20 lakh

5. Major Effort and Cost Estimation Techniques

The important techniques are:

1. **Expert Judgment**
2. **Estimation by Analogy**
3. **Top-Down Estimation**
4. **Bottom-Up Estimation**
5. **Work Breakdown Structure (WBS)**
6. **Parametric/Algorithmic Models**
7. **COCOMO**
8. **Function Point-Based Estimation**
9. **Three-Point / PERT Estimation**
10. **Price-to-Win Estimation**

Let's understand each one.

6. Expert Judgment

Definition

In **Expert Judgment**, experienced professionals estimate the effort and cost based on their knowledge and previous experience.

Experts may include:

- Project managers
- Senior developers
- Software architects
- Technical leads

Example

A project manager has previously developed three similar e-commerce applications.

Based on that experience, they estimate:

Effort = 15 person-months

Duration = 5 months

Cost = ₹12 lakh

Advantages

- Simple.
- Fast.
- Useful when historical data is limited.
- Uses practical experience.

Disadvantages

- Subjective.
 - Different experts may give different estimates.
 - Highly dependent on expert experience.
 - Personal bias can affect estimation.
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7. Estimation by Analogy

Definition

In this technique, the current project is compared with a **previously completed similar project**.

Example

Previous project:

College Management System

- Size: 25 KLOC
- Effort: 18 person-months
- Cost: ₹14 lakh

New project:

University Management System

It is slightly larger and more complex.

The manager may estimate:

- Effort $\approx$ 25 person-months
- Cost $\approx$ ₹20 lakh

Advantages

- Easy to understand.
- Uses real project data.
- Often more reliable than pure guessing.

Disadvantages

- Requires similar historical projects.
 - Projects may not actually be comparable.
 - Differences in technology and team skills can affect accuracy.
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8. Top-Down Estimation

Definition

In **Top-Down estimation**, the project is estimated **as a whole first**, and then the total effort/cost is distributed among different components.

Example

Estimated total project effort:

100 person-months

The manager distributes it:

Module	Estimated Effort
User Management	15 PM
Product Management	25 PM
Payment	20 PM
Reports	15 PM
Testing	15 PM
Deployment	10 PM
Total	100 PM

Advantages

- Fast.
- Useful during early project planning.
- Good when detailed requirements are unavailable.

Disadvantages

- Individual tasks may be underestimated.
 - Less accurate for complex projects.
 - Depends heavily on overall project experience.
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9. Bottom-Up Estimation

Definition

In **Bottom-Up estimation**, the project is divided into small tasks. Each task is estimated separately, and the estimates are added together.

Example

For an online shopping system:

Task	Effort
Login	2 PM
Product Catalog	4 PM
Shopping Cart	3 PM
Payment	5 PM
Testing	4 PM
Deployment	2 PM
Total	20 PM

Therefore:

Total estimated effort = 20 person-months

Advantages

- More detailed.
- Often more accurate.
- Helps identify individual resource requirements.
- Useful for scheduling.

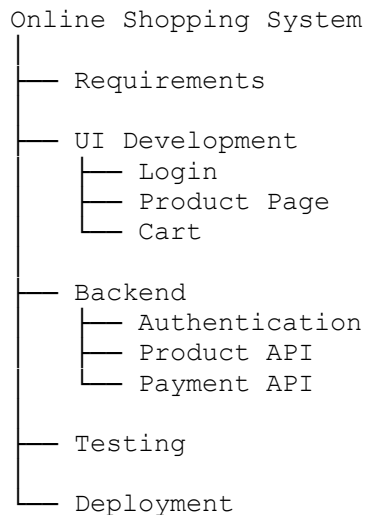
Disadvantages

- Time-consuming.
 - Requires detailed requirements.
 - Some small tasks may be forgotten.
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10. Work Breakdown Structure (WBS)

WBS is a technique for breaking a large project into smaller manageable components.

Example



Each work package is estimated separately.

Then:

Total Effort = $\sum$ Effort of all tasks

WBS is particularly useful with **bottom-up estimation**.

11. Parametric / Algorithmic Estimation

Definition

An **algorithmic model** uses mathematical equations based on measurable project characteristics.

General idea:

Effort = $f(\text{Software Size, Complexity, Other Factors})$

One of the most famous algorithmic models is **COCOMO**.

12. COCOMO Model

COCOMO stands for:

Constructive Cost Model

It was developed by **Barry Boehm**.

COCOMO estimates:

- Effort
- Development time
- Cost

based mainly on software size, usually measured in **KLOC**.

Basic COCOMO Formula

$$E = a(KLOC)^b$$

Where:

- **E** = Effort in person-months
- **KLOC** = Thousand Lines of Code
- **a, b** = constants depending on project type

Development time can be estimated using:

$$T = c(E)^d$$

Where:

- **T** = Development time in months
- **E** = Effort
- **c, d** = model constants

Then staffing can be approximated as:

$$\text{Average Staff} = TE$$

13. COCOMO Project Types

Basic COCOMO classifies projects into three categories.

1. Organic

Suitable for:

- Small projects
- Experienced teams
- Familiar technology
- Relatively simple requirements

Example:

Small payroll system.

2. Semi-Detached

Suitable for projects with:

- Medium complexity
- Mixed team experience
- Moderate size

Example:

Medium-scale business application.

3. Embedded

Suitable for:

- Highly complex systems
- Strict hardware/software constraints
- High reliability requirements

Example:

Aircraft control system.

14. Function Point Estimation

Definition

Function Point Analysis (FPA) estimates software size based on the **functionality delivered to users**, rather than lines of code.

It considers five major components:

1. **External Inputs (EI)**
2. **External Outputs (EO)**
3. **External Inquiries (EQ)**
4. **Internal Logical Files (ILF)**
5. **External Interface Files (EIF)**

Example: Banking System

Inputs:

- Deposit amount
- Withdrawal request

Outputs:

- Account statement
- Transaction report

Inquiries:

- Check balance

Internal files:

- Customer database

External interfaces:

- External payment system

These are assigned weights based on complexity to calculate function points.

15. Advantages of Function Point Estimation

- Independent of programming language.
- Can be used early in the project.
- Measures user functionality.
- Useful when LOC cannot be estimated reliably.

Disadvantages

- Requires experience.
 - Counting can be subjective.
 - More difficult for beginners.
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16. Three-Point Estimation

Software estimates contain uncertainty.

Instead of making only one estimate, three estimates are made:

Optimistic (O)

Best-case scenario.

Most Likely (M)

Most realistic estimate.

Pessimistic (P)

Worst-case scenario.

A commonly used PERT estimate is:

$$E=6O+4M+P$$

Example

Suppose development of a module is estimated as:

- O = 5 days
- M = 8 days
- P = 14 days

Then:

$$E=6(5)+4(8)+14 \quad E=65 \quad E=8.5 \text{ days}$$

So expected effort/duration is approximately **8.5 days**.

17. Price-to-Win Estimation

This is slightly different from estimating the actual development cost.

Here, the project price is determined based on **what the customer or market is willing to pay**.

Example

The actual estimated development cost is:

₹15 lakh

But the customer says:

"Our maximum budget is ₹12 lakh."

The company may decide whether it can deliver within ₹12 lakh.

Risk

The company may win the contract but make very little profit or even a loss.

18. Cost Estimation Components

When calculating total project cost, consider:

1. Labour Cost

Developer and staff salaries.

2. Hardware Cost

Servers, computers, networking equipment.

3. Software Cost

Licenses, development tools, databases, etc.

4. Cloud Cost

Hosting, storage, databases, APIs, etc.

5. Training Cost

Training employees on new technology.

6. Testing Cost

Testing tools, testers, environments.

7. Communication Cost

Internet, meetings, collaboration tools.

8. Contingency

Additional reserve for unexpected problems.

19. Example of Complete Cost Estimation

Suppose:

Development Team

5 developers $\times$ ₹70,000/month $\times$ 6 months

=₹21,00,000

Testing

=₹3,00,000

Cloud and Hardware

=₹2,00,000

Software Licenses

=₹1,00,000

Contingency

=₹3,00,000

Therefore:

Total Cost=21+3+2+1+3 =₹30 lakh

20. Effort vs Cost Estimation

Effort Estimation	Cost Estimation
Measures human work	Measures money required
Person-hours/days/months	Rupees/dollars

Effort Estimation	Cost Estimation
Focuses on labour	Includes labour + other expenses
Example: 30 person-months	Example: ₹25 lakh
Used for staffing and planning	Used for budgeting and pricing

21. Comparison of Techniques

Technique	Main Idea	Advantage	Limitation
Expert Judgment	Expert experience	Quick	Subjective
Analogy	Compare previous projects	Practical	Needs similar data
Top-Down	Estimate whole project	Fast	Less detailed
Bottom-Up	Estimate individual tasks	Detailed	Time-consuming
WBS	Break project into work packages	Organized	Requires detailed planning
COCOMO	Mathematical model	Systematic	Depends on reliable size estimate
Function Points	Estimate functionality	Language-independent	Requires expertise
Three-Point	O, M, P estimates	Handles uncertainty	More calculation
Price-to-Win	Based on customer budget	Competitive	May reduce profit

22. Important Factors Affecting Estimation

Even a good technique can produce an inaccurate estimate if important factors are ignored.

Technical Factors

- Complexity
- Technology
- Integration requirements
- Performance requirements

Human Factors

- Team experience
- Developer productivity
- Team size
- Communication

Project Factors

- Requirement stability
- Schedule constraints
- Quality requirements
- Project size

External Factors

- Market conditions
 - Vendor dependencies
 - Regulations
 - Customer availability
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23. Why Software Estimation Is Difficult

Software estimation is difficult because:

Requirements may change

Customer requirements can evolve.

Software is intangible

Unlike construction, it is difficult to physically measure software progress.

Productivity differs

Two developers may take different amounts of time for the same task.

Technology may be new

New frameworks or tools create uncertainty.

Hidden work exists

Integration, debugging, documentation, meetings, deployment, and rework may be underestimated.

Therefore:

Software estimation should be treated as an evolving prediction, not an exact number.

24. Best Practice: Use More Than One Technique

In real projects, managers often combine techniques.

For example:

WBS + Expert Judgment + Analogy + Three-Point Estimation

This gives a more reliable estimate than depending on a single method.

Example

For a payment module:

- Expert estimate = 10 days
- Historical project = 12 days
- Three-point estimate = 11 days

The manager reviews all three and produces a final estimate based on project context.

25. Easy Flow to Remember

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Understand Requirements
↓
Estimate Software Size
↓
Select Estimation Technique
↓
Estimate Effort
↓
Estimate Resources
↓
Estimate Schedule
↓
Calculate Labour Cost
↓
Add Other Costs
↓
Add Risk/Contingency
↓
Final Project Cost
```